1) a)

Extension = Current Length − Natural Length

$x = 1.35 - 1.2 = 0.15$ m

1) b)

Extension = Current Length − Natural Length

$x = 1.5 - 1.2 = 0.3$ m

1) c)

Extension = Current Length − Natural Length

$x = 1.1 - 1.2 = -0.1$ m

A negative extension means the string is not stretched. An elastic string cannot push, so it is slack.

So the extension is taken as 0 m

2) a)

$T = \frac{\lambda}{l}x$

$\lambda = 24$ N, $l = 0.8$ m, $x = 1 - 0.8 = 0.2$ m

$\therefore T = \frac{24}{0.8} \times 0.2 = 6$ N

2) b)

$T = \frac{\lambda}{l}x$

$\lambda = 24$ N, $l = 0.8$ m, $x = 1.3 - 0.8 = 0.5$ m

$\therefore T = \frac{24}{0.8} \times 0.5 = 15$ N

2) c)

$T = \frac{\lambda}{l}x$

$\lambda = 24$ N, $l = 0.8$ m, $x = 0.75 - 0.8 = -0.05$ m

The string is slack so there is no tension.

3)

$T = \frac{\lambda}{l}x$

$T = 8$ N, $l = 2$ m, $x = 2.5 - 2 = 0.5$ m

$\therefore 8 = \frac{\lambda}{2} \times 0.5 \Rightarrow \lambda = 32$ N

4)

$T = \frac{\lambda}{l}x$

$T = 4.8$ N, $l = 1.5$ m, $\lambda = 18$ N

$\therefore 4.8 = \frac{18}{1.5} \times x \Rightarrow x = 0.4$ m

Hence the length of the string is $1.5 + 0.4 = 1.9$ m

5)

$R(\uparrow): T - mg = 0$

$\Rightarrow T = 0.5 \times 9.8 = 4.9$ N

$T = \frac{\lambda}{l}x$

$\Rightarrow 4.9 = \frac{30}{1.2} \times x$

$\Rightarrow x = \frac{49}{250} = 0.196$ m

6) a)

$R(\uparrow): T - mg = 0$

$\Rightarrow T = 0.8 \times 9.8 = 7.84$ N

$T = \frac{\lambda}{l}x$

$\Rightarrow 7.84 = \frac{19.6}{0.6} \times x$

$\Rightarrow x = \frac{6}{25} = 0.24$ m

6) b)

Total Length $= 0.6 + 0.24 = 0.84$ m

7) a)

The string has natural length 1 m and its current length is y, so the extension x is equal to $(y-1)$

$T = \frac{\lambda}{l}x$

$\Rightarrow T = \frac{40}{1} \times (y-1)$

$\Rightarrow T = 40(y-1)$ as required

7) b)

If $y < 1$, then the string is shorter than its natural length, and so it is slack.

Hence $T = 0$ N

8) a)

Extension $x = y - l$

8) b)

$T = \frac{\lambda}{l}(y-l) = \lambda\frac{y-l}{l}$

8) c)

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

Taking downwards as positive, so weight mg acts downwards and is positive, while tension T acts upwards and is negative

$\Rightarrow mg - T = m\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = mg - T$

$\Rightarrow m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = mg - \lambda\frac{y-l}{l}$

9) a)

At equilibrium the spring has stretched by e beyond its natural length l, so the length of the spring is $l + e$

9) b)

If the particle moves a further distance x downwards, the spring becomes longer by x. Therefore the new length is $l + e + x$

9) c)

Extension = Current Length − Natural Length

$= (l + e + x) - l$

$= e + x$

9) d)

Hooke's Law states $T = \lambda \frac{\text{extension}}{l}$

Since the extension is $e + x$ we must have that:

$T = \lambda \frac{e+x}{l}$

9) e)

When the particle is at equilibrium, $x = 0$ and the particle's acceleration is zero. Therefore the resultant force is zero. The weight of the particle mg acts downwards, and the tension $\lambda \frac{e}{l}$ acts upwards. So:

$mg - \lambda \frac{e+0}{l} = 0$

$\Rightarrow mg = \lambda \frac{e}{l}$

10) a)

Using Hooke's Law:

$T = \lambda \frac{x}{l}$

At equilibrium the extension is e, so

$T = \lambda \frac{e}{l}$

10) b)

In equilibrium the particle has no acceleration, so the resultant force is zero.
The weight mg acts downwards, while the tension $\lambda\frac{e}{l}$ acts upwards.

$mg - \lambda\frac{e}{l} = 0$

$\Rightarrow mg = \lambda\frac{e}{l}$

10) c)

At equilibrium the extension is e. If the particle moves a further distance x downwards, the spring becomes longer by x.
So the total extension is $e + x$

10) d)

Using Hooke's Law:

$T = \lambda\frac{\text{extension}}{l}$

So:

$T = \lambda\frac{e+x}{l}$

10) e)

Since x is defined as the downward displacement from equilibrium, $x > 0$ means the particle is **below** its equilibrium position. The spring is stretched more than it is in equilibrium, so the tension acts **upwards**, towards the equilibrium position. Hence the tension makes a negative contribution to the equation of motion.

10) f)

Resultant Force $F = ma$

$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Taking downwards as positive, so weight mg acts downwards and is positive, while tension T acts upwards and is negative

$\Rightarrow mg - T = m\frac{d^2x}{dt^2}$

$\Rightarrow m\frac{d^2x}{dt^2} = mg - \lambda\frac{e+x}{l}$

10) g)

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = mg - \lambda\frac{e+x}{l}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = mg - \lambda\frac{e}{l} - \lambda\frac{x}{l}$

But at equilibrium we have $mg = \lambda\frac{e}{l}$ so:

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\lambda\frac{x}{l}$

11)

Previously, we have seen that:

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\lambda\frac{x}{l}$

Substitute $m = 0.4$, $\lambda = 8$, $l = 0.5$:

$0.4\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -8 \times \frac{x}{0.5}$

$\Rightarrow 0.4\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -16x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -40x$

12) a)

Previously, we have seen that:

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\lambda\frac{x}{l}$

Substitute $m = 0.5$, $\lambda = 20$, $l = 0.4$:

$0.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -20 \times \frac{x}{0.4}$

$\Rightarrow 0.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -50x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -100x$

12) b)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 100x = 0$

Auxiliary Equation:

$m^2 + 100 = 0$

$\Rightarrow m^2 = -100$

$\Rightarrow m = \pm 10\mathrm{i}$

General Solution:

$x = A\cos 10t + B\sin 10t$

Substitute $x = 0.1$ and $t = 0$:

$0.1 = A\cos 0 + B\sin 0$

$\Rightarrow A = 0.1$

$\therefore x = 0.1\cos 10t + B\sin 10t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin 10t + 10B\cos 10t$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -\sin 0 + 10B\cos 0$

$\Rightarrow 10B = 0$

$\Rightarrow B = 0$

$\therefore x = 0.1\cos 10t$

13) a)

Previously, we have seen that:

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\lambda\frac{x}{l}$

Substitute $m = 0.3$, $\lambda = 12$, $l = 0.5$:

$0.3\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -12 \times \frac{x}{0.5}$

$\Rightarrow 0.3\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -24x$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -80x$

13) b)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 80x = 0$

Auxiliary Equation:

$m^2 + 80 = 0$

$\Rightarrow m^2 = -80$

$\Rightarrow m = \pm 4\sqrt{5}\mathrm{i}$

General Solution:

$x = A\cos\left(4\sqrt{5}t\right) + B\sin\left(4\sqrt{5}t\right)$

Substitute $x = 0.08$ and $t = 0$:

$0.08 = A\cos 0 + B\sin 0$

$\Rightarrow A = 0.08$

$\therefore x = 0.08\cos\left(4\sqrt{5}t\right) + B\sin\left(4\sqrt{5}t\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{8\sqrt{5}}{25}\sin\left(4\sqrt{5}t\right) + 4\sqrt{5}B\cos\left(4\sqrt{5}t\right)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -\frac{8\sqrt{5}}{25}\sin 0 + 4\sqrt{5}B\cos 0$

$\Rightarrow 4\sqrt{5}B = 0$

$\Rightarrow B = 0$

$\therefore x = 0.08\cos\left(4\sqrt{5}t\right)$

14) a)

Previously, we have seen that:

$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\lambda\frac{x}{l}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{\lambda}{ml}x$

This is simple harmonic motion with $\omega = \sqrt{\frac{\lambda}{ml}}$

14) b)

Period $T = \frac{2\pi}{\omega}$

$= \frac{2\pi}{\sqrt{\frac{\lambda}{ml}}}$

$= 2\pi\sqrt{\frac{ml}{\lambda}}$

15)

The force is always directed towards O. So if the particle is on the positive side of O, the force acts in the negative direction. And if the particle is on the negative side of O, the force acts in the positive direction. Therefore, the force is always opposite in sign to the displacement.

The force has magnitude proportional to x

Hence:

$F = -kx$

where k is a positive constant.

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x$

This is of the form $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x$ where $\omega = \sqrt{\frac{k}{m}}$ which is simple harmonic motion.

16) a)

Take towards the wall as positive.

The spring is compressed by a distance x m

The spring force has magnitude $T = kx$ and opposes the compression, and so acts in the negative direction.

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -kx = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x$

This is of the form $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x$ where $\omega = \sqrt{\frac{k}{m}}$ which is simple harmonic motion.

16) b)

The period $T = \frac{2\pi}{\omega}$

$= \frac{2\pi}{\sqrt{\frac{k}{m}}}$

$= 2\pi\sqrt{\frac{m}{k}}$

16) c)

The particle oscillates about the position where the spring is neither compressed nor stretched, which is where $x = 0$

$T = k \times 0 = 0$ at this point, so there is no resultant force, making it the equilibrium position.

17) a)

$AB = 1$ m so the midpoint O is 0.5 m from each end.

Each spring has a natural length of 0.4 m.

When the particle is at O, the length of each spring is 0.5 m, and hence each spring has extension $0.5 - 0.4 = 0.1$ m

If the particle was displaced a distance x towards B, then the left-hand spring has length $(0.5 + x)$ m and the right-hand spring has length $(0.5 - x)$ m.

So the extension of the left-hand spring is $(0.5 + x) - 0.4 = 0.1 + x$

And the extension of the right-hand spring is $(0.5 - x) - 0.4 = 0.1 - x$

So the tension in the left-hand spring is $c(0.1 + x)$ and the tension in the right-hand spring is $c(0.1 - x)$

The left-hand spring pulls the particle towards A, so this force is negative, and the right-hand spring pulls the particle towards B, so this force is positive.

Resultant Force

$= c(0.1 - x) - c(0.1 + x)$

$= 0.1c - cx - 0.1c - cx$

$= -2cx$ N

17) b)

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -2cx = M\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{2c}{M}x$

This is of the form $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2x$ where $\omega = \sqrt{\frac{2c}{M}}$ which is simple harmonic motion.

17) c)

The period $T = \frac{2\pi}{\omega}$

$= \frac{2\pi}{\sqrt{\frac{2c}{M}}}$

$= 2\pi\sqrt{\frac{M}{2c}}$

18) a)

$PQ = 1$ m so the midpoint M is 0.5 m from each end.

Each spring has a natural length of 0.3 m.

When the particle is at O, the length of each spring is 0.5 m, and hence each spring has extension $0.5 - 0.3 = 0.2$ m

If the particle is displaced a distance x from M, then the left-hand spring has length $(0.5 + x)$ m and the right-hand spring has length $(0.5 - x)$ m.

So the extension of the left-hand spring is $(0.5 + x) - 0.3 = 0.2 + x$

And the extension of the right-hand spring is $(0.5 - x) - 0.3 = 0.2 - x$

So the tension in the left-hand spring is $160(0.2 + x)$ and the tension in the right-hand spring is $160(0.2 - x)$

The left-hand spring pulls the particle towards P, so this force is negative, and the right-hand spring pulls the particle towards Q, so this force is positive.

Resultant Force

$= 160(0.2 - x) - 160(0.2 + x)$

$= 32 - 160x - 32 - 160x$

$= -320x$ N

18) b)

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -320x = 8\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -40x$

This is of the form $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x$ where $\omega = \sqrt{40} = 2\sqrt{10}$ which is simple harmonic motion.

18) c)

The period $T = \frac{2\pi}{\omega}$

$= \frac{2\pi}{2\sqrt{10}}$

$= \frac{\pi}{\sqrt{10}}$

$= \frac{\pi\sqrt{10}}{10}$

18) d)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -40x$

$\Rightarrow x = A\cos\left(2\sqrt{10}t\right) + B\sin\left(2\sqrt{10}t\right)$

Substitute $t = 0$ and $x = 0.1$:

$0.1 = A\cos 0 + B\sin 0$

$\Rightarrow A = 0.1$

$\therefore x = 0.1\cos\left(2\sqrt{10}t\right) + B\sin\left(2\sqrt{10}t\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\sqrt{10}}{5}\sin\left(2\sqrt{10}t\right) + 2\sqrt{10}B\cos\left(2\sqrt{10}t\right)$

Substitute $t = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$:

$0 = -\frac{\sqrt{10}}{5}\sin 0 + 2\sqrt{10}B\cos 0$

$\Rightarrow 0 = 2\sqrt{10}B$

$\Rightarrow B = 0$

$\therefore x = 0.1\cos\left(2\sqrt{10}t\right)$

18) e)

$0.04 = 0.1\cos\left(2\sqrt{10}t\right)$

$\Rightarrow \cos\left(2\sqrt{10}t\right) = 0.4$

$\Rightarrow t = \frac{\arccos 0.4}{2\sqrt{10}} = 0.183298180\ldots$

$v = \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\sqrt{10}}{5}\sin\left(2\sqrt{10}t\right)$

$v|_{t=0.183298180\ldots} = -\frac{\sqrt{210}}{25} = -0.579655\ldots$ ms^{-1}

So the speed is $\frac{\sqrt{210}}{25} = 0.580$ ms^{-1} (3 SF)

Alternatively:

Using $v^2 = \omega^2(a^2 - x^2)$

$\Rightarrow v^2 = 40(0.1^2 - 0.04^2)$

$= \frac{42}{125}$

$\Rightarrow v = \sqrt{\frac{42}{125}} = \frac{\sqrt{210}}{25} = 0.580$ ms^{-1} (3 SF)

19) a)

At equilibrium, the extension is $0.5 - 0.3 = 0.2$ m

Since the bob is in equilibrium, the tension balances the weight, so

$T = mg = 1 \times 9.8 = 9.8$ N

We are told that the tension is proportional to the extension, so $T = ke$ for some constant k

$\Rightarrow 9.8 = k \times 0.2$

$\Rightarrow k = 49$

$\therefore T = 49e$

19) b)

At time t, the bob is x m below P, so the length of the spring is x

As the natural length is 0.3 m, the extension is $(x - 0.3)$

$\therefore T = 49(x - 0.3)$

If we take downwards as positive, we have the weight of the bob acting downwards (positive), the tension acting upwards (negative), and the resistive force opposing the direction of motion (negative).

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow 9.8 - 49(x - 0.3) - 14v = 1 \times \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 9.8 - 49x + 14.7 - 14\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 14\frac{\mathrm{d}x}{\mathrm{d}t} + 49x = 24.5$

19) c)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 14\frac{\mathrm{d}x}{\mathrm{d}t} + 49x = 24.5$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 14\frac{\mathrm{d}x}{\mathrm{d}t} + 49x = 0$

Auxiliary Equation:

$m^2 + 14m + 49 = 0$

$\Rightarrow (m+7)^2 = 0$

$\Rightarrow m = -7$

Complementary Function:

$x = At\mathrm{e}^{-7t} + B\mathrm{e}^{-7t}$

Try this Particular Integral:

$x = c$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 14\frac{\mathrm{d}x}{\mathrm{d}t} + 49x = 24.5$

$\Rightarrow 0 + 14(0) + 49(c) = 24.5$

$\Rightarrow 49c = 24.5$

$\Rightarrow c = 0.5$

Particular Integral:

$x = 0.5$

General Solution:

$x = At\mathrm{e}^{-7t} + B\mathrm{e}^{-7t} + 0.5$

Substitute $x = 0.8$ and $t = 0$:

$0.8 = A \times 0 \times \mathrm{e}^0 + B\mathrm{e}^0 + 0.5$

$\Rightarrow B = 0.3$

$\therefore x = At\mathrm{e}^{-7t} + 0.3\mathrm{e}^{-7t} + 0.5$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -7At\mathrm{e}^{-7t} + A\mathrm{e}^{-7t} - 2.1\mathrm{e}^{-7t}$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -7A \times 0 \times \mathrm{e}^0 + A\mathrm{e}^0 - 2.1\mathrm{e}^0$

$\Rightarrow 0 = A - 2.1$

$\Rightarrow A = 2.1$

$\therefore x = 2.1t\mathrm{e}^{-7t} + 0.3\mathrm{e}^{-7t} + 0.5$

19) d)

$x|_{t=0.8} = 2.1 \times 0.8 \times \mathrm{e}^{-7\times0.8} + 0.3\mathrm{e}^{-7\times0.8} + 0.5$

$= 1.68\mathrm{e}^{-5.6} + 0.3\mathrm{e}^{-5.6} + 0.5$

$= 1.98\mathrm{e}^{-5.6} + 0.5$

$= 0.50732 \ldots$

$= 0.51$ (2 SF)

The equilibrium position is $x = 0.5$ m

0.51 is close to 0.5, so the bob has almost reached its equilibrium position.

19) e)

$x = 2.1t\mathrm{e}^{-7t} + 0.3\mathrm{e}^{-7t} + 0.5$

As $t \to \infty$, $\mathrm{e}^{-7t} \to 0$

$t\mathrm{e}^{-7t} \to 0$ as e^{-7t} tends to zero faster than t tends to infinity.

So $x \to 0 + 0 + 0.5 = 0.5$

So the bob approaches $x = 0.5$ and does not oscillate.

20) a)

At time t, the particle has moved x m downwards from its initial position. So the extension of the spring is now $(0.25 + x)$

Hence the tension is $T = 16me = 16m(0.25 + x)$

If we take downwards as positive, we have the weight of the particle acting downwards (positive), the tension acting upwards (negative), and the resistive force opposing the direction of motion (negative).

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$

$mg = 9.8m$

$\Rightarrow 9.8m - 16m(0.25 + x) - 8mv = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 9.8 - 16(0.25 + x) - 8v = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 9.8 + 16(0.25 + x) + 8v = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} - 9.8 + 4 + 16x + 8\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\frac{\mathrm{d}x}{\mathrm{d}t} + 16x = 5.8$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\frac{\mathrm{d}x}{\mathrm{d}t} + 16x = 0$

Auxiliary Equation:

$m^2 + 8m + 16 = 0$

$\Rightarrow (m + 4)^2 = 0$

$\Rightarrow m = -4$

Complementary Function:

$x = Ate^{-4t} + Be^{-4t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\frac{\mathrm{d}x}{\mathrm{d}t} + 16x = 5.8$

Try this Particular Integral:

$x = c$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$\Rightarrow 0 + 8(0) + 16(c) = 5.8$

$\Rightarrow 16c = 5.8$

$\Rightarrow c = \frac{29}{80}$

Particular Integral:

$x = \frac{29}{80}$

General Solution:

$x = Ate^{-4t} + Be^{-4t} + \frac{29}{80}$

Substitute $x = 0$ and $t = 0$:

$0 = A \times 0 \times e^{0} + Be^{0} + \frac{29}{80}$

$\Rightarrow 0 = B + \frac{29}{80}$

$\Rightarrow B = -\frac{29}{80}$

$\therefore x = Ate^{-4t} - \frac{29}{80}e^{-4t} + \frac{29}{80}$

$\frac{dx}{dt} = -4Ate^{-4t} + Ae^{-4t} + \frac{29}{20}e^{-4t} + \frac{29}{80}$

Substitute $\frac{dx}{dt} = 0$ and $t = 0$:

$0 = -4A \times 0 \times e^{0} + Ae^{0} + \frac{29}{20}e^{0} + \frac{29}{80}$

$\Rightarrow 0 = A + \frac{29}{20}$

$\Rightarrow A = -\frac{29}{20}$

$\therefore x = -\frac{29}{20}te^{-4t} - \frac{29}{80}e^{-4t} + \frac{29}{80}$

20) b)

The auxiliary equation has one repeated real root, so the motion is critically damped.

21) a)

The standard form is $T = ke$ where k is the stiffness (measured in N m^{-1}) and e is the extension (measured in m). So in this case, the stiffness is 6 N m^{-1}

21) b)

Natural Length = 2 m

Current Length = 2.3 m

So the extension is $2.3 - 2 = 0.3$ m

$\therefore T = 6 \times 0.3 = 1.8$ N

21) c)

We are given $T = 9$ N

$\therefore 9 = 6e$

$\Rightarrow e = \frac{3}{2}$

So the length of the string is $2 + \frac{3}{2} = 3.5$ m

22) a)

Let the equilibrium position of the slider be E, and let x be the displacement of the slider from E, where displacement towards R is taken as positive.

$ER + LE = 10 \Rightarrow ER = 10 - LE$

The extension of the cord LS is $(LE - 3)$

The extension of the cord SR is $(ER - 2) = \big((10 - LE) - 2\big) = (8 - LE)$

The tension in the left-hand coord is $2d(LE - 3)$

The tension in the right-hand coord is $6d(8 - LE)$

At equilibrium, these tensions are equal:

$2d(LE - 3) = 6d(8 - LE)$

$\Rightarrow LE - 3 = 3(8 - LE)$

$\Rightarrow LE - 3 = 24 - 3LE$

$\Rightarrow 4LE = 27$

$\Rightarrow LE = \frac{27}{4} = 6.75$ m

So the equilibrium position is 6.75 m from L

Now consider the slider being displaced a distance x m to the right of equilibrium.

Then the length of the left-hand cord becomes $\left(\frac{27}{4} + x\right)$

So its extension is $\left(\left(\frac{27}{4} + x\right) - 3\right) = \left(\frac{15}{4} + x\right)$

And the tension is $2d\left(\frac{15}{4} + x\right)$

The length of the right-hand cord becomes $\left(10 - \left(\frac{27}{4} + x\right)\right) = \left(\frac{13}{4} - x\right)$

So its extension is $\left(\left(\frac{13}{4} - x\right) - 2\right) = \left(\frac{5}{4} - x\right)$

And the tension is $6d\left(\frac{5}{4} - x\right)$

The left hand cord pulls to the left (negative), and the right-hand cord pulls to the right (positive).

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow 6d\left(\frac{5}{4} - x\right) - 2d\left(\frac{15}{4} + x\right) = 4d\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{3}{2}\left(\frac{5}{4} - x\right) - \frac{1}{2}\left(\frac{15}{4} + x\right) = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{15}{8} - \frac{3}{2}x - \frac{15}{8} - \frac{1}{2}x = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -2x$

This is of the form $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2x$ where $\omega = \sqrt{2}$ which is simple harmonic motion.

22) b)

Angular Frequency $= \omega = \sqrt{2}$

22) c)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -2x$

$\Rightarrow x = A\cos\sqrt{2}t + B\sin\sqrt{2}t$

Substitute $x = 0.2$ and $t = 0$:

$0.2 = A\cos 0 + B\sin 0$

$\Rightarrow A = 0.2$

$\therefore x = 0.2\cos\sqrt{2}t + B\sin\sqrt{2}t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\sqrt{2}}{5}\sin\sqrt{2}t + B\sqrt{2}\cos\sqrt{2}t$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -\frac{\sqrt{2}}{5}\sin 0 + B\sqrt{2}\cos 0$

$\Rightarrow B = 0$

$\therefore x = 0.2\cos\sqrt{2}t$

$0.12 = 0.2\cos\sqrt{2}t$

$\Rightarrow \cos\sqrt{2}t = \frac{3}{5}$

$\Rightarrow t = \frac{\arccos\left(\frac{3}{5}\right)}{\sqrt{2}} = 0.6556967\ldots$

$v = \frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\sqrt{2}}{5}\sin\left(\sqrt{2}t\right)$

$\left.\frac{\mathrm{d}x}{\mathrm{d}t}\right| = -\frac{\sqrt{2}}{5}\sin\left(\sqrt{2}\times\frac{\arccos\left(\frac{3}{5}\right)}{\sqrt{2}}\right)$

$= -\frac{4\sqrt{2}}{25}$ ms-1

Speed is $\frac{4\sqrt{2}}{25}$ ms^{-1} $= 0.226$ ms^{-1}

Alternatively:

Using $v^2 = \omega^2(a^2 - x^2)$

$\Rightarrow v^2 = 2(0.2^2 - 0.12^2)$

$= \frac{32}{625}$

$\Rightarrow v = \sqrt{\frac{32}{625}} = \frac{4\sqrt{2}}{25} = 0.226$ ms^{-1} (3 SF)

23) a)

If $LQ = 2.4$ m, then since $LR = 5$ m, we must have $QR = 5 - 2.4 = 2.6$ m

For string LQ:

Natural length $= 1.2$ m, actual length $= 2.4$ m, so extension $= 2.4 - 1.2 = 1.2$ m

Hence tension in LQ is $18me = 18m \times 1.2 = 21.6m$ N

For string QR:

Natural length $= 1.8$ m, actual length $= 2.6$ m, so extension $= 2.6 - 1.8 = 0.8$ m

Hence tension in QR is $27me = 27m \times 0.8 = 21.6m$ N

So the tension in LQ is equal to the tension in QR

23) b)

We are told that at time t, $LQ = 2.4 + x$

As $LR = 5$ m, we must have that $QR = 5 - (2.4 + x) = 2.6 - x$

For string LQ:

Natural length $= 1.2$ m, actual length $= (2.4 + x)$ m, so extension

$= (2.4 + x) - 1.2 = (1.2 + x)$ m

Hence tension in LQ is $18me = 18m \times (1.2 + x) = (21.6m + 18mx)$ N

For string QR:

Natural length $= 1.8$ m, actual length $= (2.6 - x)$ m, so extension

$= (2.6 - x) - 1.8 = (0.8 - x)$ m

Hence tension in QR is $27me = 27m \times (0.8 - x) = (21.6m - 27mx)$ N

Friction:

$R(\uparrow)$: $R - mg = 0 \Rightarrow R = mg$

$\therefore Fr = \mu R = 0.5m \times 9.8 = 4.9m$

We will take the positive direction to be towards R. So tension in LQ is acting to the left (negative), tension in QR is acting to the right (positive), and friction is acting to the right (positive) as the block is moving to the left.

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow (21.6m - 27mx) + 4.9m - (21.6m + 18mx) = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 4.9m - 45mx = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 45mx = 4.9m$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 45x = 4.9$

23) c)

There is no longer friction on the shelf, but there is a resistive force of magnitude $8mv$ N opposing the motion.

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$ so $8mv = 8m\frac{\mathrm{d}x}{\mathrm{d}t}$

Resultant Force $F = ma$

$\Rightarrow (21.6m - 27mx) - (21.6m + 18mx) - 8m\frac{\mathrm{d}x}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -45mx - 8m\frac{\mathrm{d}x}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8m\frac{\mathrm{d}x}{\mathrm{d}t} + 45mx = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\frac{\mathrm{d}x}{\mathrm{d}t} + 45x = 0$

Auxiliary Equation:

$m^2 + 8m + 45 = 0$

$\Rightarrow (m+4)^2 - 16 + 45 = 0$

$\Rightarrow (m+4)^2 = -29$

$\Rightarrow m = -4 \pm \sqrt{29}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-4t}\left(A\cos\left(\sqrt{29}t\right) + B\sin\left(\sqrt{29}t\right)\right)$

Note that the block is released from the point where $LQ = 2.8$, but as $LQ = 2.4 + x$ we must have that $x = 0.4$

Substitute $x = 0.4$ and $t = 0$:

$0.4 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow A = 0.4$

$\therefore x = \mathrm{e}^{-4t}\left(0.4\cos\left(\sqrt{29}t\right) + B\sin\left(\sqrt{29}t\right)\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-4t}\left(-\frac{2\sqrt{29}}{5}\sin\left(\sqrt{29}t\right) + B\sqrt{29}\cos\left(\sqrt{29}t\right)\right) - 4\mathrm{e}^{-4t}\left(0.4\cos\left(\sqrt{29}t\right) + B\sin\left(\sqrt{29}t\right)\right)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^0\left(-\frac{2\sqrt{29}}{5}\sin 0 + B\sqrt{29}\cos 0\right) - 4\mathrm{e}^0(0.4\cos 0 + B\sin 0)$

$\Rightarrow 0 = B\sqrt{29} - 1.6$

$\Rightarrow B = \frac{1.6}{\sqrt{29}} = \frac{8\sqrt{29}}{145}$

$\therefore x = \mathrm{e}^{-4t}\left(0.4\cos\left(\sqrt{29}t\right) + \frac{8\sqrt{29}}{145}\sin\left(\sqrt{29}t\right)\right)$

24) a)

The particle moves a distance x down the slope.

The extension of the upper string will be $a + x$

The extension of the lower string will be $b - x$

24) b)

$mg\sin\theta$

24) c)

The tension in the upper string is $T = k_1 e = k_1(a + x)$ acting up the slope.

The tension in the lower string is $T = k_2 e = k_2(b - x)$ acting down the slope.

The component of weight down the slope is $mg\sin\theta$

Resultant Force $F = ma$

$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

$\Rightarrow k_2(b - x) + mg\sin\theta - k_1(a + x) = m\frac{d^2x}{dt^2}$

25) a)

Let the equilibrium position of the bead be E so that $ME = p$

Then $EN = 4.2 - p$

When the bead is displaced x m down the wire from equilibrium, the length of the upper string becomes $(p + x)$ m, and the length of the lower string becomes $(4.2 - p - x)$ m

Extension in upper string $= (p + x - 1.5)$ m

Extension in lower string $= 4.2 - p - x - 0.9 = (3.3 - p - x)$ m

So:

The tension in the upper string is $T = 8e = 8(p + x - 1.5)$ N

The tension in the lower string is $T = 12e = 12(3.3 - p - x)$ N

We will take the positive direction to be down the slope. So tension in the lower string is acting down the slope (positive), tension in the upper string is acting up the slope (negative), and the component of weight parallel to the slope acts down the slope (positive).

Resultant Force $F = ma$

$a = \dot{v} = \ddot{x}$

$\Rightarrow 12(3.3 - p - x) + 0.5 \times 9.8 \times \sin 30° - 8(p + x - 1.5) = 0.5\ddot{x}$

$\Rightarrow 39.6 - 12p - 12x + 2.45 - 8p - 8x + 12 = 0.5\ddot{x}$

When the bead is at equilibrium, then $x = 0$ and $\ddot{x} = 0$, so:

$\Rightarrow 39.6 - 12p + 2.45 - 8p + 12 = 0$

$\Rightarrow 20p = 54.05$

$\Rightarrow p = \frac{1081}{400}$

Then:

$39.6 - 12 \times \frac{1081}{400} - 12x + 2.45 - 8 \times \frac{1081}{400} - 8x + 12 = 0.5\ddot{x}$

$\Rightarrow 39.6 - 32.43 - 12x + 2.45 - 21.62 - 8x + 12 = 0.5\ddot{x}$

$\Rightarrow -20x = 0.5\ddot{x}$

$\Rightarrow -40x = \ddot{x}$

$\Rightarrow \ddot{x} = -40x$

This is of the form $\ddot{x} = -\omega^2 x$ with $\omega^2 = 40 \Rightarrow \omega = 2\sqrt{10}$

25) b)

With a resistive force added, the equation becomes:

$-20x - c\dot{x} = 0.5\ddot{x}$

$\Rightarrow 0.5\ddot{x} + c\dot{x} + 20x = 0$

$\Rightarrow \ddot{x} + 2c\dot{x} + 40x = 0$

Auxiliary Equation:

$m^2 + 2cm + 40 = 0$

$\Rightarrow (m + c)^2 - c^2 + 40 = 0$

$\Rightarrow (m + c)^2 = c^2 - 40$

$\Rightarrow m = -c \pm \sqrt{c^2 - 40}$

The motion is critically damped so $c^2 - 40 = 0$

$\Rightarrow c = \sqrt{40} = 2\sqrt{10}$ as $c > 0$

25) c)

$m = -2\sqrt{10}$

General Solution:

$x = Ate^{-2\sqrt{10}t} + Be^{-2\sqrt{10}t}$

The bead is released from rest from a point 0.15 m below equilibrium, so substitute $x = 0.15$ and $t = 0$:

$0.15 = A \times 0 \times e^0 + Be^0$

$\Rightarrow 0.15 = B$

$\therefore x = Ate^{-2\sqrt{10}t} + 0.15e^{-2\sqrt{10}t}$

$\dot{x} = -2\sqrt{10}Ate^{-2\sqrt{10}t} + Ae^{-2\sqrt{10}t} - \frac{3\sqrt{10}}{10}e^{-2\sqrt{10}t}$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = -2\sqrt{10}A \times 0 \times e^0 + Ae^0 - \frac{3\sqrt{10}}{10}e^0$

$\Rightarrow 0 = A - \frac{3\sqrt{10}}{10}$

$\Rightarrow A = \frac{3\sqrt{10}}{10}$

$\therefore x = \frac{3\sqrt{10}}{10}te^{-2\sqrt{10}t} + \frac{3}{20}e^{-2\sqrt{10}t}$

25) d)

$x = \frac{3\sqrt{10}}{10} te^{-2\sqrt{10}t} + \frac{3}{20} e^{-2\sqrt{10}t}$

$v = \dot{x} = -6te^{-2\sqrt{10}t} + \frac{3\sqrt{10}}{10} e^{-2\sqrt{10}t} - \frac{3\sqrt{10}}{10} e^{-2\sqrt{10}t}$

$\Rightarrow v = -6te^{-2\sqrt{10}t}$

$a = \dot{v} = \ddot{x} = 12\sqrt{10}te^{-2\sqrt{10}t} - 6e^{-2\sqrt{10}t}$

Maximum speed occurs when $a = 0$:

$12\sqrt{10}te^{-2\sqrt{10}t} - 6e^{-2\sqrt{10}t} = 0$

$\Rightarrow 2\sqrt{10}te^{-2\sqrt{10}t} - e^{-2\sqrt{10}t} = 0$

$\Rightarrow \left(2\sqrt{10}t - 1\right)e^{-2\sqrt{10}t} = 0$

$\Rightarrow 2\sqrt{10}t - 1 = 0$

$\Rightarrow t = \frac{1}{2\sqrt{10}} = \frac{\sqrt{10}}{20}$

$v|_{t=\frac{\sqrt{10}}{20}} = -6 \times \frac{\sqrt{10}}{20} \times e^{-2\sqrt{10} \times \frac{\sqrt{10}}{20}}$

$= -\frac{3\sqrt{10}}{10} e^{-1}$

So maximum speed is

$\frac{3\sqrt{10}}{10e}$ ms^{-1}

26) a)

Let e be the equilibrium extension of the string.

The component of the weight down the slope is $mg\sin\theta = 1 \times 10 \times \sin 45° = 5\sqrt{2}$ N

The tension in the string is $T = \frac{\lambda}{l} e = \frac{20}{2} e = 10e$

In equilibrium, the tension and the component of the weight down the slope balance:

$10e = 5\sqrt{2}$

$\Rightarrow e = \frac{\sqrt{2}}{2}$ m

26) b)

At time t, the extension of the string is $(e + x)$

The tension in the string is $T = \frac{20}{2}(e + x) = 10(e + x)$

Take down the slope as positive, so the component of the weight acts down the slope (positive), the tension acts up the slope (negative), and resistive forces oppose the motion (negative).

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$

$\Rightarrow 5\sqrt{2} - 10(e + x) - 2v = 1 \times \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 5\sqrt{2} - 10e - 10x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 5\sqrt{2} - 10 \times \frac{\sqrt{2}}{2} - 10x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -10x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 10x = 0$

26) c)

Auxiliary Equation:

$m^2 + 2m + 10 = 0$

$\Rightarrow (m+1)^2 - 1 + 10 = 0$

$\Rightarrow (m+1)^2 = -9$

$\Rightarrow m = -1 \pm 3\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-t}(A\cos 3t + B\sin 3t)$

Substitute $x = 0.2$ and $t = 0$:

$0.2 = \mathrm{e}^{0}(A\cos 0 + B\sin 0)$

$\Rightarrow 0.2 = A$

$\Rightarrow A = \frac{1}{5}$

$\therefore x = \mathrm{e}^{-t}\left(\frac{1}{5}\cos 3t + B\sin 3t\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-t}\left(-\frac{3}{5}\sin 3t + 3B\cos 3t\right) - \mathrm{e}^{-t}\left(\frac{1}{5}\cos 3t + B\sin 3t\right)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}\left(-\frac{3}{5}\sin 0 + 3B\cos 0\right) - \mathrm{e}^{0}\left(\frac{1}{5}\cos 0 + B\sin 0\right)$

$\Rightarrow 0 = 3B - \frac{1}{5}$

$\Rightarrow B = \frac{1}{15}$

$\therefore x = \mathrm{e}^{-t}\left(\frac{1}{5}\cos 3t + \frac{1}{15}\sin 3t\right)$

26) d)

$x = \mathrm{e}^{-t}\left(\frac{1}{5}\cos 3t + \frac{1}{15}\sin 3t\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-t}\left(-\frac{3}{5}\sin 3t + \frac{1}{5}\cos 3t\right) - \mathrm{e}^{-t}\left(\frac{1}{5}\cos 3t + \frac{1}{15}\sin 3t\right)$

$= \mathrm{e}^{-t}\left(-\frac{3}{5}\sin 3t + \frac{1}{5}\cos 3t - \frac{1}{5}\cos 3t - \frac{1}{15}\sin 3t\right)$

$= \mathrm{e}^{-t}\left(-\frac{2}{3}\sin 3t\right)$

$= -\frac{2}{3}\mathrm{e}^{-t}\sin 3t$

For the particle to be momentarily at rest, $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\Rightarrow -\frac{2}{3}\mathrm{e}^{-t}\sin 3t = 0$

$\Rightarrow \sin 3t = 0$

The first time after release is when $3t = \pi \Rightarrow t = \frac{\pi}{3}$

So:

$x|_{t=\frac{\pi}{3}} = \mathrm{e}^{-\frac{\pi}{3}}\left(\frac{1}{5}\cos\left(3 \times \frac{\pi}{3}\right) + \frac{1}{15}\sin\left(3 \times \frac{\pi}{3}\right)\right)$

$= \mathrm{e}^{-\frac{\pi}{3}}\left(\frac{1}{5}\cos\pi + \frac{1}{15}\sin\pi\right)$

$= -\frac{1}{5}\mathrm{e}^{-\frac{\pi}{3}}$

So at the first turning point after release, the particle is $\frac{1}{5}\mathrm{e}^{-\frac{\pi}{3}}$ m **up the slope** from equilibrium.